

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 8

GREATEST AMONG TWO NUMBERS

AIM: To write and execute an 8085 assembly language program to find greatest among two numbers and store the result in memory.

ALGORITHM

1. Load the first number into register A.
2. Load the second number into register B.
3. Compare the two numbers using CMP.
4. If A is greater than or equal to B, store A as the greatest.
5. Otherwise, store B as the greatest.
6. Stop the program.

CODE:

```
MVI A,25H
MVI B,35H
CMP B
JNC STORE_A
MOV A,B
STORE_A: STA 8050H
HLT
```

OUTPUT:

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	22
BC	19 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 10
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

Load me at

```
1 LDA 2500
2 MOV B,A
3
4 LDA 2501
5 CMP B
6 JNC STORE
7
8 MOV A,B
9
10 STORE: STA 2502
11 HLT
```

Decimal - Hex Conversion

Decimal	Hex
0	0

To Hex To Dec

I/O Ports

Port	Value
0	00

Update Port Value

Memory

Address	Value
2501	22

Update Memory

Start 2500

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	34
09C6	2502	34
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

```
0 Program assembled successfully
```

Simulator: Idle

Windows Taskbar: Search, File Explorer, Edge, VS Code, HP, WhatsApp, Chrome, 11:40 28-07-2026